Lance Rivera

Greater Sydney Area, NSW | Australian Permanent Resident
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Education

University of New South Wales Feb 2022 - Dec 2026
Bachelor of Science (Computer Science) *Sydney, NSW*
Relevant Coursework: Data Structures and Algorithms, Computer Systems Fundamentals, Discrete Mathematics

Skills

Languages: TypeScript, JavaScript, Python, C
Backend & Data: Node.js, Fastify, PostgreSQL
Cloud & DevOps: AWS (Lightsail, ECR, Lambda, SQS, S3, CloudFront), Docker, AWS SAM, GitHub Actions, Claude Code
Frontend: React, Next.js, Tailwind CSS

Experience

Junior Systems Engineer Mar 2026 - Present
Pool Safety Solutions *Sydney, NSW*

- Productionised a locally built franchise management prototype for a network operating across NSW, QLD and VIC. Reviewed the existing codebase and fixed security issues, including missing tenant isolation.
- Reworked the PostgreSQL data layer for multi-tenancy, adding row-level security so franchise data is isolated by tenant. Encoded business rules tied to Franchising Code 2024 requirements as database constraints instead of manual checks.
- Set up separate staging and production environments and deployed containerised services with Docker on AWS using Lightsail, ECR, S3 and CloudFront. Built a Lambda/SQS ingestion pipeline with AWS SAM to sync GoHighLevel data into PostgreSQL.
- Connected GoHighLevel, Stripe and Xero integrations through APIs and webhooks, covering authentication, data synchronisation and failure handling.
- Built and maintain about 470 automated tests and a GitHub Actions CI/CD pipeline that gates changes through staging before production. Improved CI speed with Docker layer caching and parallel jobs;
- Use AI coding agents (Claude Code) for spec-driven development with review gates, running parallel agents on separate git worktrees to speed up delivery without compromising review quality.

Administrative Assistant, Full-time National Service Nov 2022 - Nov 2024
Singapore Armed Forces *Singapore*

- Coordinated VBA automation for recurring data-entry work, reducing processing time by 90%.
- Built a Telegram bot with JavaScript, Google Apps Script and the Telegram API to replace a manual communication process.

Projects

Airtable Clone | Live Demo

- Built editable tables, views, filters and sorting with Next.js, TypeScript, tRPC, Prisma and PostgreSQL.
- Used pagination, memoisation and virtualisation to keep tables responsive with more than 1 million rows.

Cloud-Hosted Personal Website | lanceriver.com

- Built on serverless AWS infrastructure with S3 static hosting, API Gateway, DynamoDB and Python Lambda for the backend.
- Managed with AWS SAM (Infrastructure as Code) and a GitHub Actions CI/CD pipeline for automated deploys.

Extracurriculars

Marketing Subcommittee Mar 2025 - Dec 2025
UNSW Computer Science and Engineering Society *Sydney, NSW*

- Managed social content for an audience of more than 9,000 followers; posts, reels and TikTok videos reached over 50,000 cumulative views.